Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

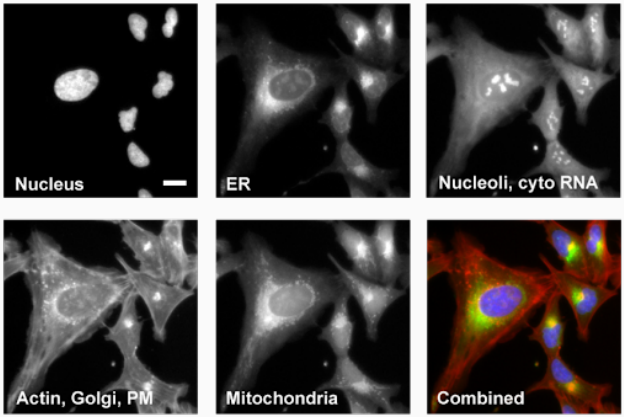
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Carpenter-Shen Lab, Purdue University; Carpenter-Singh Lab, Broad Institute

10 November 2026

What is Cell Painting?

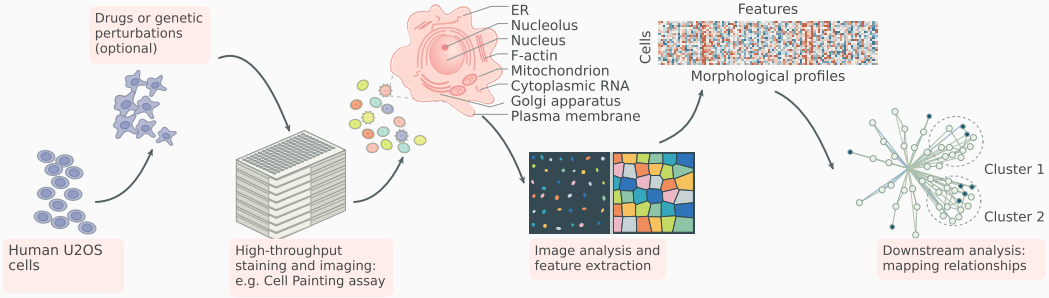
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.



Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.



Adapted from Chandrasekaran et al. (2021), Fig. 1a

JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

CRISPR knockout	Gene overexpression (ORF)	Compound treatment
7,975 gene targets	**12,609** genes	**115,794** compounds

We make JUMP accessible via the JUMP Hub

Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

How-to guides	No-code gene/compound queries; explore similarity heatmaps.
Tutorials	Curated runnable notebooks for profiles, metadata and statistical analysis.
Further resources	Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists.

Was my compound tested? Nocodazole

A familiar biological question

Can morphology recover nocodazole's annotated mechanism?

- Nocodazole is present: `JCP2022_047835`.
- Annotated mechanism: tubulin polymerization inhibitor.
- Colchicine, vinblastine, vincristine and mebendazole are also present.
- For this example, use CellProfiler-derived profiles only.

Start with a simple query: "Find nocodazole and its annotations in JUMP."

JUMP_rr compounds; Repurposing annotations; checked 9 November 2026

Which compounds look similar to nocodazole?

Query: “Show nocodazole’s closest published matches.”

Annotated peer	Similarity	JUMP reagent ID
Colchicine	0.934	JCP2022_033814
CYT-997	0.916	JCP2022_056401
Parbendazole	0.916	JCP2022_110440
Oxibendazole	0.902	JCP2022_077096

- Four annotated peers occur among the five highest returned matches.
- Similar morphology suggests a useful biological connection.

CellProfiler-derived, batch-corrected profiles; published ranked subset, not an exhaustive screen.

JUMP_rr compound matches; checked 9 November 2026

Does nocodazole change cell morphology?

Published activity result

Corrected activity p-value: **0.00003**

- A detectable profile in this published analysis.
- Fewer neighboring cells: Cohen's $d = -3.306$.
- Lower nuclear form factor: Cohen's $d = -3.110$.

Follow-up question: could cell density contribute to the matches?

JUMP_rr compound features; checked 9 November 2026

Use JUMP to investigate a phenotype

Simple questions in JUMP_rr

1. Was nocodazole tested?
2. Does it have a detectable profile?
3. Which compounds look similar?
4. Which measurements change?

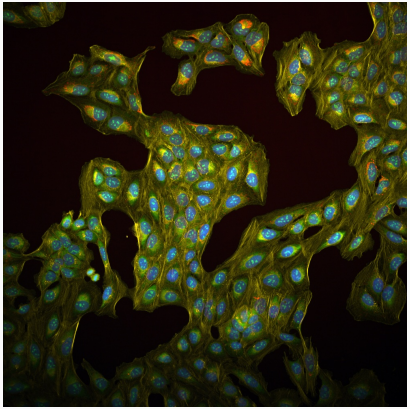
Then investigate the result

- Check images and matched controls.
- Check consistency across replicates.
- Test whether cell density explains part of the similarity.

One compound, one biological question, CellProfiler profiles.

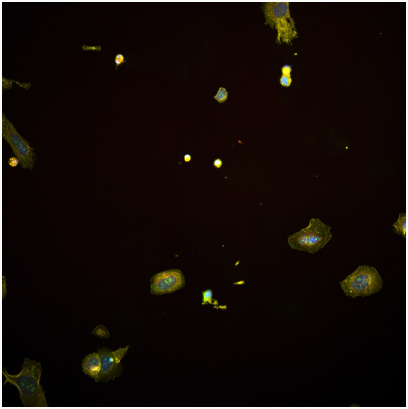
What do nocodazole-treated cells look like?

DMSO control



A02, site 1

Nocodazole



P15, site 1

Same source and plate: source_2 / 1086292020. Inspect density and cell appearance.

JUMP gallery / Ardigen previews

Live demo: investigate nocodazole's matches

From a simple query to an investigation

Investigate whether nocodazole retrieves other microtubule-disrupting compounds using CellProfiler profiles. Compare their retrieval against unrelated compounds, check consistency across replicates, and show representative images alongside the quantitative results.

My follow-up:

"Could this reflect cell density or toxicity? Check cell counts and quality-control measurements."

Finish with:

"Turn the result into one slide with an image comparison, a retrieval plot and the main limitation. Build a preview."

External agent and backends on Oppy; forwarded interface on the MacBook.

What we know, and what we will check

Checked before the demo

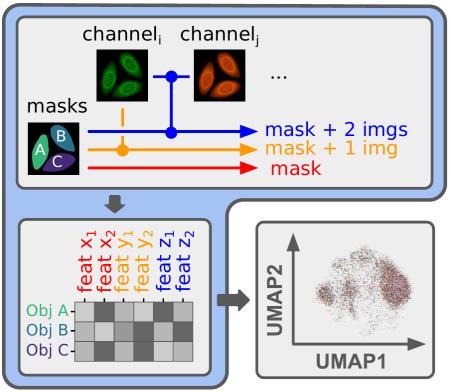
- Nocodazole and 19 annotated peers are present.
- Colchicine is its top returned match.
- Images and same-plate DMSO controls are available.

The live investigation

- How consistent are the replicates?
- Are annotated peers preferentially retrieved?
- How much is explained by density or QC?

From a recognizable phenotype to a testable biological hypothesis.

Not just JUMP: Extract morphological profiles from your own images



cp_measure: feature extraction in Python

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

Applications of JUMP and our tools across biological questions

Evaluate new lysosomal fluorescence probes

Kenneth Li, Vendrell Lab, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes/imaging methods are better for an different Mechanisms of Action

MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

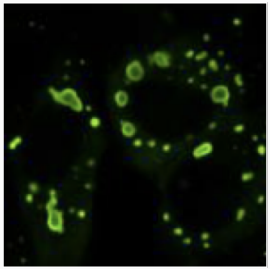
- Integration of morphological features with Squidpy.

MisLocus: a single-cell image benchmark

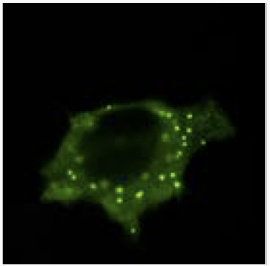
MisLocus contains ∼3 million cells spanning 1,541 reference/variant alleles across 282 genes, excluding controls.

Does a variant change where the protein is located?

LZTR1 reference



LZTR1 G248R



Unpublished data (preprint coming soon)

Protein mislocalisation

- Compare reference and variant proteins.
- Test for localisation changes.
- Evaluate image representations.

Images from Taipale lab

Take-home messages

- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

Thank you

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JUMP partners

JUMP supporting partners

JUMP collaborators

Scientific:
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a PerkinElmer Company
Administrative:
Pistoia Alliance
Academic:
Umeå University
Stanford Machine Learning Group

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References

All named tools and models below are clickable.

Assay, data and resources

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.
- Demo recording; Target-2 artifacts.
- word2vec (Mikolov, 2013).

Image-analysis toolbox

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- Nahual: model interfaces.
- marimo; uv.
- Squidpy; Morpheus.

Representations and methods

- DINOv2; DINOv3.
- MorphEm.
- OpenPhenom.
- SubCell.
- ViT-random: DINOv2 code.
- Harmony; JUMP recipe.
- Parquet; Zarr; JPEG XL.

JUMP Hub and slides

JUMP Hub



broad.io/jump

Slides (PDF)



Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

Classical features (e.g., CellProfiler)

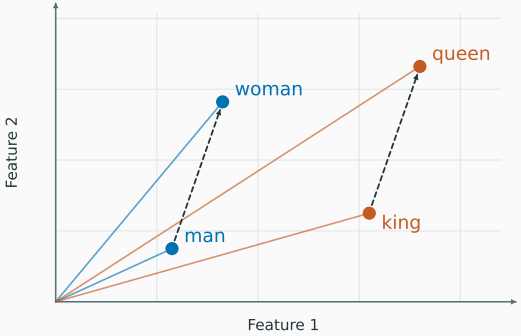
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

Learned representations

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

Words as vectors: word2vec

Learning can be viewed as compression



Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

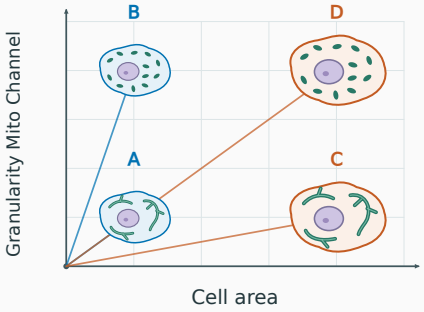
Mikolov et al. (2013)

From words to cells: vectors for comparison

Classical features

Defined measurements

x = [cell area, granularity, ...]

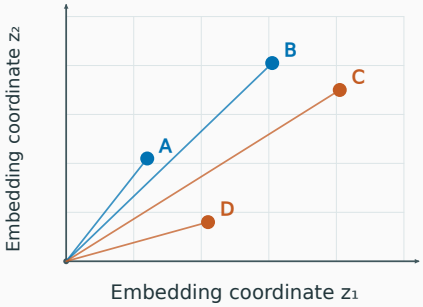


Learned embeddings

Learned from images

Not necessarily directly interpretable.

z = [z_1, z_2, ..., z_n]



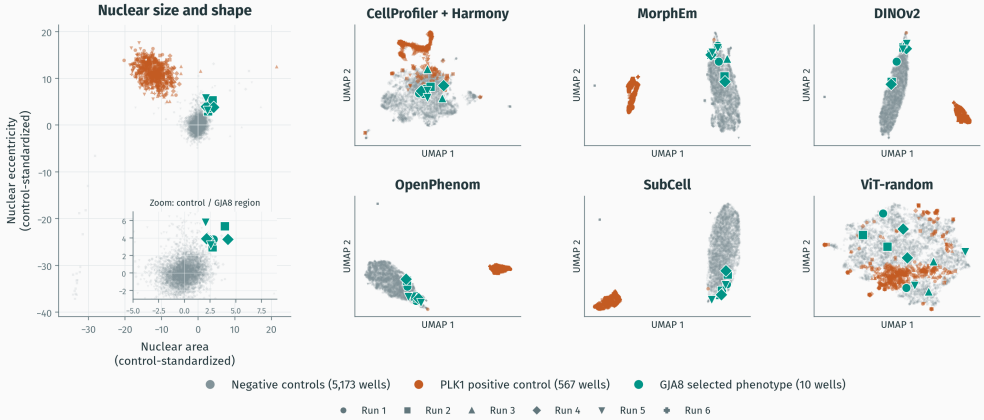
Control

Perturbation

Modern tools make image analysis more accessible

Find cells, nuclei and spots	Cellpose (cells and nuclei); Spotiflow (spots)
Follow cells and organelles over time	Trackastra / Ultrack
Measure cell and organelle appearance	cp_measure
Explore your results	marimo

Positive controls contextualize perturbations in both learned and classical representations



Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

JUMP-lite: a compact benchmark

Same images. Different methods.



Test whether profiles recover known relationships.



Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

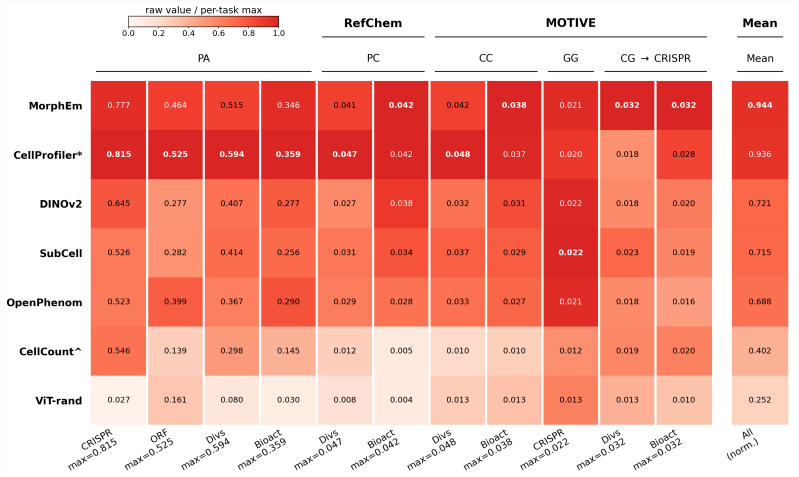
Nahual: swap models within one workflow

Swap models. Keep your workflow.

Image features	DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm
Segmentation	Cellpose ↔ StarDist
Spot detection	Spotiflow
Tracking	Trackastra ↔ Ultrack

JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.

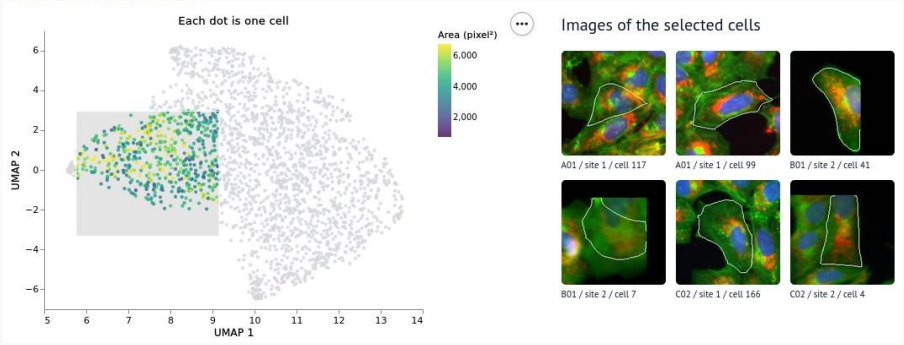


Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5

Backup: explore cell subpopulations

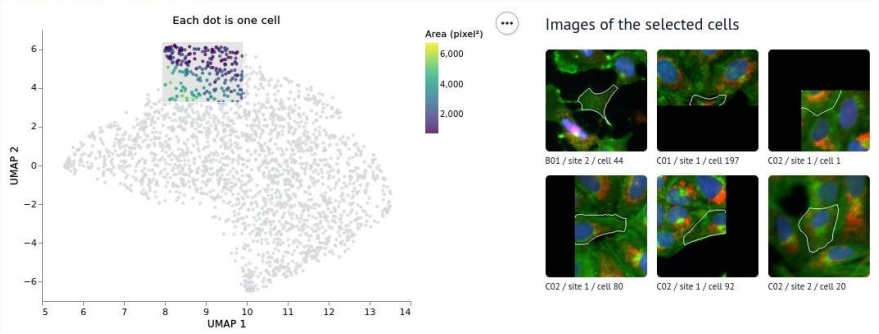
Interactive marimo notebook — separate recorded example



Each dot is a cell: 2,223 from one Target-2 plate. Shown selection: 545 cells. Inspect images, area and other features.

Source: Muñoz (2026), JUMP Target-2 notebook recording

Backup: compare another cell selection



- Second selection: 224 cells; compare images and measurements.
- Not the nocodazole query; no treatment/control assignments in this example.

Source: Muñoz (2026), JUMP Target-2 notebook recording